Mock viva question bank

40 questions tagged by person (A / B / both) and difficulty, with short model answers. Numbers refer to the current code and data.

How to use this bank

- Cover the answers, read the question aloud, answer in 30-60 seconds, then compare.
- Practise your own questions first, then 'both', then the other person's.
- **Mock viva with Claude:** in the Claude chat, type "run the mock viva for Person A" (or Person B, or both). Claude acts as the examiner: one question at a time from this bank plus sensible follow-ups. It waits for your answer, says what was correct, what was missing and what the ideal answer is, then waits for you before the next question.

	easy	medium	hard	total
A	3	5	5	13
B	3	5	5	13
both	3	5	6	14

Person A

Q1. [A · easy] What does `nn.Conv2d(3, 32, kernel_size=3, padding=1)` do?
Learns 32 filters of size 3×3 over the 3 colour channels; padding 1 keeps 224×224. Output 32×224×224.

Q2. [A · easy] What are the output shapes after each conv_block?
32×112×112, 64×56×56, 128×28×28, 256×14×14 (each MaxPool halves height and width).

Q3. [A · easy] How many parameters does SimpleCNN have?
1,183,942 (about 1.18M).

Q4. [A · medium] Why Dropout 0.3 only before the last layer?
The conv layers already have BatchNorm and few parameters; the classifier is where memorising is easiest. 0.3 is a moderate value.

Q5. [A · medium] Your fit function: does it do early stopping?
No. It runs all epochs and saves the checkpoint whenever validation accuracy improves, so we keep the best epoch, but training never stops early.

Q6. [A · medium] What is in the 'other' column of the PlantDoc confusion matrix?
Predictions of PlantVillage classes that PlantDoc does not have; they are always wrong.

Q7. [A · medium] How is a random background made?
A random 4×4 colour grid is enlarged to the image size (bicubic) and Gaussian noise (std 20) is added; a new one for every training image load (random_background in data.py).

Q8. [A · medium] Why does validation use the same background as training in the background runs?
Validation chooses the best checkpoint; it must look like the training task. The test conditions then vary the background on purpose.

Q9. [A · hard] Why are the random test backgrounds fixed with a seed?
So every model is tested on exactly the same images; otherwise differences could come from luck in the backgrounds.

Q10. [A · hard] What does a leaf-focus score equal to the mean leaf area tell you?
The heat is spread as if the model did not care where the leaf is: no evidence of leaf focus.

Q11. [A · hard] Why ReLU in Grad-CAM?
Negative values are regions that lower the class score (evidence for other classes); we only want the positive evidence.

Q12. [A · hard] Why can't you compute the leaf-focus score on PlantDoc?
PlantDoc has no leaf masks; we would need to segment the leaves ourselves.

Q13. [A · hard] cnn_standard drops on 'leaf only' test images. Is that background bias?
Partly, but careful: a black background is also new (out of distribution) for it. The random-background test and the cnn_random_bg run make the argument stronger.

Person B

Q14. [B · easy] Which weights did you load?
torchvision `EfficientNet_B0_Weights.IMAGENET1K_V1`.

Q15. [B · easy] Which layer did you replace?
`model.classifier[1]`: Linear(1280, 1000) -> Linear(1280, 38).

Q16. [B · easy] How many parameters are trainable?
All 4,056,226: no layer is frozen.

Q17. [B · medium] Why fine-tune all layers?
Leaf images differ a lot from ImageNet, we have several thousand images, and the small LR (0.0005) protects the pretrained features. (We did not run the frozen variant to compare.)

Q18. [B · medium] What is the LR in the last epoch of EfficientNet training?
About 3.3e-05 (cosine schedule, T_max = 6, stepped per epoch; see the LR table).

Q19. [B · medium] List the strong augmentations with their parameters.
RandomResizedCrop(224, scale 0.5-1), H/V flips, RandomRotation(30), ColorJitter(0.4, 0.4, 0.4, 0.05), GaussianBlur(9, sigma 0.1-3) with p=0.3.

Q20. [B · medium] Why is augmentation not applied to validation and test images?
We want to measure performance on real images as they are, and the measurement must be repeatable.

Q21. [B · medium] How is the domain gap computed?
Accuracy on PlantVillage test restricted to the 28 shared classes, minus accuracy on PlantDoc (same 28 classes).

Q22. [B · hard] What does keep_correct=0.95 mean?
The threshold is the 5th percentile of the confidence of correct validation predictions: 95% of correct lab answers stay above it.

Q23. [B · hard] Coverage on PlantDoc is much lower than on PlantVillage. Good or bad?
Expected and useful: the model is less sure on unfamiliar images and refuses more. Bad only if the accuracy on the answered images does not improve.

Q24. [B · hard] Why not use a fixed threshold like 0.9?
0.9 is arbitrary. Our rule is tied to how confident the model is when it is right on validation data, and it is chosen without seeing the test sets.

Q25. [B · hard] What is the risk of over-confident softmax outputs?
Out-of-domain images can get high confidence and pass the threshold; calibration (e.g. temperature scaling) or OOD detection would be needed.

Q26. [B · hard] Why use EfficientNet-B0's last conv block for Grad-CAM?
It is the last layer with spatial information (1280×7×7) and the most class-specific features.

Both

Q27. [both · easy] What is the input tensor shape?
3×224×224 per image, 64×3×224×224 per batch.

Q28. [both · easy] How many classes, and how many are shared with PlantDoc?
38 classes; 28 shared.

Q29. [both · easy] What are the split sizes?
7,430 train, 1,520 val, 2,244 test (PlantVillage); PlantDoc 2,578, test only.

Q30. [both · medium] How did you avoid leakage?
Leaf-grouped train/val split, the official leaf-grouped test split, PlantDoc never used in training, and the epoch and threshold chosen on validation.

Q31. [both · medium] Why macro-F1 in addition to accuracy?
Every class counts equally, so weak small classes are visible.

Q32. [both · medium] What does set_seed do?
Seeds Python, NumPy and PyTorch (CPU and GPU) and makes cuDNN deterministic, so runs repeat.

Q33. [both · medium] Why mixed precision?
float16 on the T4 is about 2× faster; GradScaler prevents tiny gradients from becoming zero.

Q34. [both · medium] Where are checkpoints saved on Colab, and why?
In MyDrive/plant_disease/checkpoints, so they survive when the free runtime disconnects.

Q35. [both · hard] Name two weaknesses of your evaluation.
Small per-class test sets and a single run per experiment (no variance); also the squashed PlantDoc images and the 2 leaf groups shared between train and test.

Q36. [both · hard] What is novel in your work?
Not the model: the systematic check of lab-to-field robustness (domain gap, background swaps, leaf focus, rejection).

Q37. [both · hard] If you had labelled field data, what would you change first?
Fine-tune on part of it, keep a separate field validation set for the threshold and the epoch, and test on a held-out field set.

Q38. [both · hard] How does the app decide to say 'NOT SURE'?
If the softmax confidence of the top class is below the threshold from results/rejection/<model>.json (default 0.5 if missing).

Q39. [both · hard] Why compare models with the same seed, data and epochs per model type?
So that the only difference is the factor we study (pretraining, augmentation or background).

Q40. [both · hard] What would show that strong augmentation really helped?
A higher PlantDoc accuracy and a smaller domain gap than the standard run of the same model; ideally repeated with several seeds.